
The Geopolitics of Oil

Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

Advanced Econometrics Project

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Dataset: Emerging Importers Panel + Full Mixed Panel

Period: January 2023 – June 2026 (42 months)

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Contents

1	Introduction	3
2	Theoretical Framework	3
2.1	Conceptual Background	3
2.2	Panel Data Specification	4
2.3	Identification Strategy	4
3	Data and Sample Construction	5
3.1	Data Sources and Variables	5
3.2	Country Samples	5
3.2.1	Primary Sample: Emerging Importers	5
3.2.2	Heterogeneity Sample: Full Mixed Panel	7
3.3	Descriptive Statistics	7
4	Econometric Methodology	9
4.1	Model Specifications	9
4.2	Diagnostic Tests	9
4.3	Standard Errors	9
5	Empirical Results	10
5.1	CPI Models: Inflation Impact	10
5.1.1	POLS Identification Check	10
5.1.2	Full Panel Estimator Comparison — CPI	10
5.2	IPI Models: Industrial Production Impact	11
5.2.1	POLS Identification Check	12
5.2.2	Full Panel Estimator Comparison — IPI	12
6	Diagnostic Tests	13
7	Heterogeneity Analysis	14
7.1	Oil Importers vs. Exporters	14
7.2	Advanced vs. Emerging Economies	15
8	Discussion and Policy Implications	16
8.1	Synthesis of Findings	16
8.2	Scenario Alignment	16
8.3	Policy Implications	17
9	Limitations	17

10 Conclusion	18
Appendix: R Code Structure Overview	18

1. Introduction

On February 27, 2026, escalating tensions in the Persian Gulf ignited a geopolitical oil shock that rapidly transmitted through global commodity markets. Brent crude prices surged sharply in March 2026, sending inflationary pressures and industrial uncertainty across both advanced and emerging economies. This event constitutes a near-textbook quasi-natural experiment: a sudden, exogenous supply-side shock whose timing and geographic origin are plausibly orthogonal to the domestic macroeconomic fundamentals of most countries in our sample.

This report applies panel data econometrics to quantify the macroeconomic damage from this shock. Specifically, we estimate the effect of the Gulf Crisis on two key outcomes — the Consumer Price Index (CPI) as a measure of inflationary pressure, and the Industrial Production Index (IPI) as a proxy for real economic activity — across a panel of emerging market economies that are net importers of oil.

Our primary estimation sample consists of five emerging importers: India, Indonesia, Poland, South Africa, and Turkey (42 monthly observations per country, January 2023 to June 2026). We complement this with a broader seven-country heterogeneity analysis covering both importers and exporters, and both advanced and emerging market economies.

The analysis proceeds in three steps. First, we establish a descriptive baseline and justify sample construction decisions. Second, we estimate Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models and select among them using Hausman and Breusch-Pagan diagnostics. Third, we conduct heterogeneity analyses to ask whether the crisis impact differed systematically by oil-balance status (importer vs. exporter) and development level (advanced vs. emerging).

2. Theoretical Framework

2.1. Conceptual Background

An oil price shock transmits to the macroeconomy through several well-documented channels. For net oil-importing nations, higher energy costs raise production costs directly (cost-push inflation) and indirectly compress household real incomes (demand compression). The combined effect is *stagflation*: simultaneous upward pressure on prices and downward pressure on output.

For net oil-exporters, the short-run effect on income is positive — higher export revenues improve the current account and fiscal position. However, geopolitical crises in the Gulf can simultaneously disrupt transportation and production infrastructure, introducing uncertainty effects that partially offset the revenue gains.

2.2. Panel Data Specification

The general panel data equation estimated in this project is:

$$y_{it} = \alpha_i + \lambda_t + \beta_1 \text{OilShock}_t + \beta_2 \ln(\text{NEER}_{it}) + \beta_3 r_{it} + \beta_4 \text{VIX}_t + \varepsilon_{it} \quad (1)$$

where the subscripts i and t index country and month respectively, and the variables are:

Table 1. Variable Definitions and Expected Signs

Symbol	Definition	CPI sign	IPI sign
y_{it}	$\ln(\text{CPI}_{it})$ or $\ln(\text{IPI}_{it})$	—	—
α_i	Country fixed effect (unobserved heterogeneity)	—	—
λ_t	Time fixed effect (common global shocks)	—	—
OilShock_t	% change in Brent crude; or <i>post_crisis</i> dummy	+	—
$\ln(\text{NEER}_{it})$	Log nominal effective exchange rate	—	+
r_{it}	Central bank policy rate	—	—
VIX_t	CBOE global risk/volatility index	+	—
ε_{it}	Idiosyncratic error	—	—

2.3. Identification Strategy

A critical econometric issue shapes our estimation strategy: the Brent crude oil price series (oil_pct_change_t) has **no cross-sectional variation** — it is identical across all countries in any given month. Under the Fixed Effects within-group transformation, any variable without cross-sectional variation is perfectly collinear with time fixed effects and cannot be separately identified.

We therefore adopt a two-track identification approach:

Identification Strategy

POLS only: Include oil_pct_change_t directly. Since Pooled OLS applies no demeaning, the variable retains its time-series dimension and captures direct oil price pass-through to CPI/IPI.

FE and RE: Replace oil_pct_change_t with a *post_crisis* dummy (= 1 for March–June 2026). This reduced-form proxy captures the structural break induced by the Gulf Crisis and is identifiable under the within-group transformation.

Robustness check: We also estimate a POLS specification that includes both oil_pct_change_t and *post_crisis* simultaneously, to verify multicollinearity and confirm that the crisis dummy absorbs the oil price coefficient.

3. Data and Sample Construction

3.1. Data Sources and Variables

All data are monthly frequency, spanning January 2023 to June 2026 (42 months). Table 2 summarises the variables and their sources.

Table 2. Variables and Data Sources

Variable	Symbol	Description	Source
CPI (log)	$\ln \text{CPI}_{it}$	Consumer Price Index (2015=100), log-transformed	IMF IFS, OECD
IPI (log)	$\ln \text{IPI}_{it}$	Industrial Production Index (2015=100), log-transformed	IMF IFS, FRED
Brent / Oil Shock	OilShock_t	% change in Brent crude price	EIA, FRED
NEER (log)	$\ln \text{NEER}_{it}$	Nominal Effective Exchange Rate, broad index	BIS, FRED
Policy Rate	r_{it}	Central bank policy rate (%)	BIS, national CBs
VIX	VIX_t	CBOE Volatility Index (global risk proxy)	FRED (VIXCLS)
Post-Crisis Dummy	post_crisis_t	= 1 for March–June 2026	Constructed

Values for April–June 2026 that were not yet available at data collection were generated using ARIMA time-series forecasting, applied country-by-country. The quality of these forecasts varies across countries and is discussed below.

3.2. Country Samples

3.2.1. Primary Sample: Emerging Importers

The project guidelines specify six emerging net-importers: India, Indonesia, Poland, South Africa, Thailand, and Turkey. Following a systematic data quality audit, two countries

were excluded from some or all regression models.

Thailand (excluded from all models): Thailand’s CPI ranged from 99.24 to 100.98 over the entire 42-month window — a total spread of less than two index points, with a within-country variance of approximately 0.18 (compared to 13.9 for India and 12.9 for Poland). The Fixed Effects estimator identifies parameters purely from within-country time variation; a near-flat series contributes no identifying information to the CPI model and risks attenuating all estimated coefficients toward zero. Thailand’s near-deflation during 2023–2025 is a well-documented structural phenomenon, not a data collection failure, and it renders Thailand uninformative for the within-group estimator. Thailand is therefore excluded from all CPI and IPI models.

South Africa (excluded from IPI models only): South Africa’s IPI is observed only through November 2023 — 11 months out of 42. The remaining 31 observations (74% of the series), spanning all of 2024, all of 2025, and the entire post-crisis period, are ARIMA-forecasted. Critically, the ARIMA model converged to a flat constant ($IPI = 110.25$) for every month from December 2023 onward, producing zero within-country variation over the most policy-relevant portion of the sample. Including this series would mean the post-crisis coefficient is effectively estimated from a constant, attenuating $\hat{\beta}$ and inflating standard errors. South Africa is retained in CPI models where data quality is acceptable, but excluded from all IPI specifications.

India (retained with caveat in IPI models): India’s IPI is observed through October 2024 (22 of 42 months). The remaining 20 months (48%) are ARIMA-forecasted, but unlike South Africa, these forecasts exhibit realistic seasonal variation rather than converging to a constant. Dropping India would leave only three countries in the IPI panel, severely limiting statistical power. India is retained but the limitation is acknowledged: the estimated post-crisis IPI effect for India is partly identified from ARIMA extrapolations of pre-crisis trends, which likely understates the true shock response.

Table 3. Data Exclusion Summary

Country	CPI Sample	IPI Sample	Reason
India	Included	Included*	*48% IPI ARIMA; acknowledged
Indonesia	Included	Included	Data quality acceptable
Poland	Included	Included	Data quality acceptable
South Africa	Included	Excluded	74% IPI ARIMA; flat constant from Dec 2023
Thailand	Excluded	Excluded	CPI variance ≈ 0.18 ; near-zero range
Turkey	Included	Included	Data quality acceptable

Final CPI panel: Indonesia, India, Poland, South Africa, Turkey — 5 countries, 210 obs

Final IPI panel: Indonesia, India, Poland, Turkey — 4 countries, 168 obs

3.2.2. Heterogeneity Sample: Full Mixed Panel

For the heterogeneity analysis we use a broader seven-country panel stratified by two dimensions. Table 4 shows the composition.

Table 4. Full Mixed Panel: Country Groups

Group	Countries	Status
Advanced Importer	Germany	CPI & IPI
Advanced Exporter	Canada, USA	CPI & IPI
Emerging Importer	India, Poland, Turkey	CPI & IPI
Emerging Exporter	Saudi Arabia	CPI only (no monthly IPI)

3.3. Descriptive Statistics

Tables 5 and 6 present summary statistics for the primary estimation samples.

Table 5. Descriptive Statistics — CPI Estimation Sample (5 Countries, 210 Obs)

Variable	N	Mean	Std Dev	Min	Max
CPI (Index)	210	119.44	26.51	92.79	237.41
Brent (USD/bbl)	210	84.37	14.88	62.10	118.25
NEER (Index)	210	91.06	13.82	53.10	108.10
Policy Rate (%)	210	11.72	13.79	0.00	50.00
VIX	210	19.84	7.23	12.10	44.20
Oil Shock (% ch.)	210	0.34	5.77	−14.61	23.91

Sample: Indonesia, India, Poland, South Africa, Turkey. Thailand excluded.

Table 6. Descriptive Statistics — IPI Estimation Sample (4 Countries, 168 Obs)

Variable	N	Mean	Std Dev	Min	Max
IPI (Index)	168	120.34	17.43	84.26	175.81
Brent (USD/bbl)	168	84.37	14.88	62.10	118.25
NEER (Index)	168	90.22	12.01	66.50	108.10
Policy Rate (%)	168	10.04	13.26	0.00	50.00
VIX	168	19.84	7.23	12.10	44.20
Oil Shock (% ch.)	168	0.34	5.77	−14.61	23.91

Sample: Indonesia, India, Poland, Turkey. Thailand and South Africa excluded.

Several features of the data are worth noting:

- **High CPI heterogeneity:** The standard deviation of CPI across the panel is 26.5 index points, primarily driven by Turkey’s prolonged inflation episode. This heterogeneity underscores the importance of country fixed effects.
- **Oil shock distribution:** Monthly Brent price changes average near zero over the full sample (0.34%) but show substantial variance (std dev 5.77%), with the March 2026 spike of approximately +24% standing as the largest single-month movement in the sample.
- **VIX spike:** The maximum VIX value (44.2) coincides with the post-crisis period, reflecting elevated global risk aversion following the Gulf escalation.
- **Policy rate dispersion:** The 50-percentage-point spread in policy rates (Turkey’s

emergency tightening vs. near-zero rates in early 2023) further motivates the use of fixed effects to control for cross-country differences.

4. Econometric Methodology

4.1. Model Specifications

We estimate five specifications for each dependent variable (CPI and IPI).

- M1. POLS (oil only):** $y_{it} = \beta_0 + \beta_1 \text{oil_pct_change}_t + \beta_2 \ln \text{NEER}_{it} + \beta_3 r_{it} + \beta_4 \text{VIX}_t + \varepsilon_{it}$
Captures direct oil price transmission. Baseline with no country heterogeneity.
- M2. POLS (oil + crisis):** Adds post_crisis_t to M1.
Robustness check for multicollinearity between the two oil shock proxies.
- M3. POLS (crisis only):** $y_{it} = \beta_0 + \beta_1 \text{post_crisis}_t + \beta_2 \ln \text{NEER}_{it} + \beta_3 r_{it} + \beta_4 \text{VIX}_t + \varepsilon_{it}$
Baseline pooled specification consistent with the FE/RE estimators.
- M4. Fixed Effects (FE):** Adds country dummies α_i to M3 (within estimator).
Controls for all time-invariant country-level unobservables.
- M5. Random Effects (RE):** Treats α_i as random draws uncorrelated with regressors.
Reported for completeness; FE is preferred on theoretical grounds.

4.2. Diagnostic Tests

Three diagnostic tests guide model selection:

- **Hausman test (FE vs. RE):** Tests whether country fixed effects are correlated with the regressors. Rejection ($p < 0.05$) favours FE. Given the small number of countries (4–5), the test may have low power; FE is retained on theoretical grounds regardless.
- **Breusch-Pagan LM test (POLS vs. RE):** Tests for the presence of country-specific random effects. Rejection indicates that Pooled OLS is insufficient and a panel model is needed.
- **Breusch-Godfrey test:** Tests for serial correlation in panel residuals. Significant autocorrelation motivates the use of heteroskedasticity and autocorrelation consistent (HAC) standard errors.

4.3. Standard Errors

All reported standard errors are robust. For POLS specifications, we use White HC3 heteroskedasticity-consistent standard errors. For FE and RE models, we use Arellano HC3 cluster-robust standard errors, which account for both heteroskedasticity and within-cluster (within-country) serial correlation.

5. Empirical Results

5.1. CPI Models: Inflation Impact

5.1.1. POLS Identification Check

Table 7 reports the three POLS specifications for $\log(\text{CPI})$, designed to verify the multicollinearity between the oil price change variable and the post-crisis dummy.

Table 7. POLS Specifications for $\text{Log}(\text{CPI})$ — Oil Shock Identification

	(1) Oil only	(2) Oil + Crisis	(3) Crisis only
Oil Shock (% Δ Brent)	+	+ (attenuated)	—
Post-Crisis Dummy	—	+	+
$\ln(\text{NEER})$	—	—	—
Policy Rate	+	+	+
VIX	+	+	+
R^2	0.71	0.73	0.72
Observations	210	210	210

HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$. Sample: Indonesia, India, Poland, South Africa, Turkey.

Col (2) confirms multicollinearity: oil shock loses significance when `post_crisis` is present.

The key finding from Table 7 is that in column (2), the oil shock coefficient loses statistical significance once the post-crisis dummy is included, while the post-crisis dummy itself becomes significant. This confirms the professor's observation that the two variables are collinear: the post-crisis dummy absorbs the oil price signal. Column (3) is therefore the appropriate POLS baseline for comparison with FE and RE.

5.1.2. Full Panel Estimator Comparison — CPI

Table 8 presents the core results across POLS, FE, and RE for the CPI outcome.

Table 8. Panel Estimators for Log(CPI)

	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects
Post-Crisis Dummy	+0.027**	+0.031**	+0.029**
ln(NEER)	−0.41**	−0.58**	−0.52**
Policy Rate	+0.003*	+0.004*	+0.003*
VIX	+0.002	+0.001	+0.001
Country FE	No	Yes	Random
R^2	0.72	0.89	0.87
Observations	210	210	210

Arellano HC3 robust SEs for FE/RE; HC3 for POLS. ** $p < 0.05$, * $p < 0.10$.

Sample: Indonesia, India, Poland, South Africa, Turkey.

Interpretation of the preferred FE model (column 2):

- **Post-Crisis Dummy (+0.031**):** The Gulf Crisis caused an average increase of approximately 3.1% in the CPI across the five emerging importers, relative to the pre-crisis baseline. This is statistically significant and consistent with the expected stagflationary effect of an oil supply shock on importing economies.
- **NEER (−0.58**):** A 1% appreciation in the nominal effective exchange rate reduces the CPI by approximately 0.58%. This aligns with theory: a stronger domestic currency lowers import prices, reducing inflationary pressure. The FE coefficient is larger in magnitude than the POLS estimate (−0.41), suggesting POLS was biased toward zero due to omitted country-level heterogeneity.
- **Policy Rate (+0.004*):** The positive sign is somewhat counterintuitive at the aggregate level but likely reflects reverse causality: countries with high inflation tend to raise rates, so the association is partly mechanical rather than structural.
- **VIX:** The VIX coefficient is positive but statistically insignificant in the FE model, suggesting that once country effects and the crisis dummy are controlled for, global financial volatility has limited additional explanatory power for CPI.
- **R^2 jump from POLS to FE (0.72 → 0.89)** confirms that country fixed effects explain a substantial share of the variation in log(CPI), validating the use of the within estimator.

5.2. IPI Models: Industrial Production Impact

5.2.1. POLS Identification Check

The multicollinearity pattern observed for CPI replicates for IPI: the oil shock coefficient in the joint POLS specification loses significance once the post-crisis dummy is included, and the crisis dummy absorbs the oil price signal.

Table 9. POLS Specifications for Log(IPI) — Oil Shock Identification

	(1) Oil only	(2) Oil + Crisis	(3) Crisis only
Oil Shock (% Δ Brent)	—*	— (attenuated)	—
Post-Crisis Dummy	—	—*	—**
ln(NEER)	+**	+**	+**
Policy Rate	—*	—*	—*
VIX	—*	—	—*
R^2	0.38	0.41	0.40
Observations	168	168	168

HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$. Sample: Indonesia, India, Poland, Turkey.

5.2.2. Full Panel Estimator Comparison — IPI

Table 10. Panel Estimators for Log(IPI)

	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects
Post-Crisis Dummy	−0.018**	−0.022**	−0.020**
ln(NEER)	+0.31**	+0.47**	+0.39**
Policy Rate	−0.002*	−0.003*	−0.002*
VIX	−0.002*	−0.002*	−0.002*
Country FE	No	Yes	Random
R^2	0.40	0.76	0.72
Observations	168	168	168

Arellano HC3 robust SEs for FE/RE; HC3 for POLS. ** $p < 0.05$, * $p < 0.10$.

Sample: Indonesia, India, Poland, Turkey. India: 48% ARIMA IPI — see limitation.

Interpretation of the preferred FE model (column 2):

- **Post-Crisis Dummy** (-0.022^{**}): The Gulf Crisis caused an average *decline* of approximately 2.2% in industrial production across the four emerging importers. This is the stagflationary supply-side effect in the real economy: higher energy input costs compress manufacturing output.
- **NEER** ($+0.47^{**}$): Exchange rate appreciation is associated with higher industrial production. A stronger currency reduces the cost of imported capital goods and intermediate inputs, supporting output. The FE estimate again exceeds the POLS estimate, consistent with omitted variable bias correction.
- **Policy Rate** (-0.003^*): Higher policy rates reduce industrial production, consistent with contractionary monetary policy dampening investment and credit-financed output.
- **VIX** (-0.002^*): Global uncertainty reduces industrial production, consistent with the risk aversion / investment suppression channel.
- **R^2 jump** ($0.40 \rightarrow 0.76$) is even more dramatic for IPI than for CPI, underscoring the importance of controlling for country-level industrial structure.

6. Diagnostic Tests

Table 11 summarises the three diagnostic tests applied to both outcomes.

Table 11. Diagnostic Test Results

Test	Null Hypothesis	CPI	IPI
Hausman (FE vs. RE)	RE consistent (no correlation with α_i)	$p < 0.05$	$p > 0.05$
<i>Decision:</i>	<i>CPI: FE preferred (H_0 rejected). IPI: RE acceptable, but FE retained on theoretical grounds.</i>		
Breusch-Pagan LM	No individual random effects (POLS sufficient)	$p < 0.001$	$p < 0.001$
<i>Decision:</i>	<i>Both: POLS rejected. Panel model (FE or RE) is needed.</i>		
Breusch-Godfrey	No serial correlation in residuals	$p < 0.05$	$p < 0.05$
<i>Decision:</i>	<i>Both: Serial correlation present. Arellano HC3 robust SEs warranted (applied throughout).</i>		

Hausman test discussion: The Hausman test rejects the null for CPI, providing formal statistical support for FE over RE. For IPI, the test fails to reject; however, given the small cross-section ($N = 4$ countries), this is likely a power problem rather than genuine evidence of RE consistency. Theoretically, country-level fixed effects — capturing industrial structure, energy intensity, import dependence, and institutional quality — are almost

certainly correlated with the NEER and policy rate regressors. FE is therefore retained as the preferred specification for both outcomes.

7. Heterogeneity Analysis

7.1. Oil Importers vs. Exporters

Table 12. CPI Heterogeneity: Importers vs. Exporters (FE, Robust SE)

	Oil Importers	Oil Exporters
	Germany, India, Poland, Turkey	Canada, Saudi Arabia, USA
Post-Crisis Dummy	+0.033**	+0.008
ln(NEER)	−0.54**	−0.21*
Policy Rate	+0.003*	−0.001
VIX	+0.001	−0.002*
R^2	0.88	0.71
Observations	168	126

Arellano HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$.

Table 13. IPI Heterogeneity: Importers vs. Exporters (FE, Robust SE)

	Oil Importers	Oil Exporters
	Germany, India, Poland, Turkey	Canada, USA (Saudi Arabia excluded)
Post-Crisis Dummy	−0.026**	+0.011
ln(NEER)	+0.44**	+0.18*
Policy Rate	−0.003*	−0.001
VIX	−0.002*	+0.001
R^2	0.77	0.68
Observations	168	84

Saudi Arabia excluded (no monthly IPI data). Arellano HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$.

The importer–exporter split reveals a stark asymmetry. Oil importers show a statistically significant positive effect on CPI (+3.3%) and a negative effect on IPI (−2.6%) — the stagflationary pattern predicted by theory. Oil exporters, by contrast, show a small positive CPI effect that is not statistically significant (+0.8%), and a slightly positive (also insignificant) IPI effect. This is consistent with the export revenue windfall partially offsetting the inflationary impact for resource-abundant economies.

7.2. Advanced vs. Emerging Economies

Table 14. CPI Heterogeneity: Advanced vs. Emerging (FE, Robust SE)

	Advanced Economies	Emerging Economies
	Germany, Canada, USA	India, Poland, Turkey, Saudi Arabia
Post-Crisis Dummy	+0.014*	+0.041**
ln(NEER)	−0.19*	−0.63**
Policy Rate	−0.001	+0.004*
VIX	+0.001	+0.002
R^2	0.74	0.91
Observations	126	168

Arellano HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$.

Table 15. IPI Heterogeneity: Advanced vs. Emerging (FE, Robust SE)

	Advanced Economies	Emerging Economies
	Germany, Canada, USA	India, Poland, Turkey
Post-Crisis Dummy	−0.011	−0.031**
ln(NEER)	+0.22*	+0.53**
Policy Rate	−0.001	−0.004**
VIX	−0.001	−0.003*
R^2	0.69	0.80
Observations	126	126

Saudi Arabia excluded. Arellano HC3 robust SEs. ** $p < 0.05$, * $p < 0.10$.

The advanced–emerging comparison yields equally important findings. Emerging economies experience a much larger inflationary pass-through from the crisis (+4.1% vs. +1.4%) and a significantly larger output contraction (−3.1% vs. −1.1%, the latter insignificant). This is consistent with structural vulnerabilities in emerging markets: weaker monetary credibility, higher energy intensity in production, and greater exchange rate pass-through to consumer prices. Advanced economies appear better insulated from oil shocks, likely because of deeper energy efficiency, more credible inflation expectations anchoring, and more diversified industrial bases.

8. Discussion and Policy Implications

8.1. Synthesis of Findings

Our results paint a coherent picture of the macroeconomic transmission of the 2026 Gulf oil shock:

1. **Stagflation confirmed:** The primary FE models document a simultaneous increase in CPI (+3.1%) and decrease in IPI (−2.2%) among emerging importing economies following the crisis onset. This constitutes a textbook supply-side stagflationary shock.
2. **Exchange rate is a key buffer:** The NEER coefficient is large and consistently significant across all specifications. Countries whose currencies appreciated following the oil shock experienced lower inflationary pressure and a smaller output decline, confirming the exchange rate as a natural shock absorber.
3. **Heterogeneity is economically meaningful:** The crisis impact is not homogeneous across countries. Emerging importers bear the brunt of both the inflation and the output contraction; advanced economies and oil exporters are comparatively insulated.
4. **Fixed effects are necessary:** The R^2 jumps from 0.72 to 0.89 (CPI) and 0.40 to 0.76 (IPI) when country fixed effects are added, and Breusch-Pagan tests reject POLS. This confirms that unobserved, time-invariant country characteristics — energy intensity, institutional quality, industrial structure — are critical determinants of macroeconomic outcomes.

8.2. Scenario Alignment

The project’s scenario framework distinguishes between a brief supply disruption (Scenario 1), prolonged uncertainty (Scenario 2), and acute geopolitical escalation (Scenario 3). Our estimated post-crisis CPI effect of approximately 3% over four months is consistent with a Scenario 2 trajectory. The absence of a Scenario 3-level response (which would imply sustained double-digit oil price spikes and potential stagflationary spirals) may reflect the short post-crisis window available in the data (March–June 2026, only four months), which

likely under-captures second-round effects.

8.3. Policy Implications

- **Exchange rate management:** Allowing or facilitating currency appreciation in the aftermath of an oil shock can meaningfully dampen inflationary pass-through, as evidenced by the large, significant NEER coefficient.
- **Monetary policy trade-off:** Higher policy rates reduce industrial output, creating a difficult trade-off for central banks facing simultaneous inflation and output decline. The negative IPI-policy rate coefficient highlights this tension.
- **Differential vulnerability:** International policy coordination should recognise that emerging importers are disproportionately affected and may need targeted support (IMF facilities, currency swap lines, energy subsidy fiscal space) relative to advanced economies.

9. Limitations

Data and sample limitations:

1. **Thailand exclusion:** Thailand's near-zero CPI variance renders it uninformative for the within estimator. This is not a data error but a structural feature of Thailand's economy during this period, and it limits the representativeness of the Southeast Asian component of the panel.
2. **South Africa IPI:** With 74% of the IPI series ARIMA-forecasted and converging to a flat constant, South Africa's industrial production data is effectively unusable for the post-crisis identification. The exclusion is justified but reduces the African country representation to zero in IPI models.
3. **India ARIMA:** India's post-crisis IPI is entirely derived from ARIMA extrapolation of pre-crisis trends. This mechanically projects forward the pre-crisis trajectory and likely understates the true shock response, biasing the estimated post-crisis IPI coefficient for India toward zero.
4. **Short post-crisis window:** Only four months of post-crisis data are available (March–June 2026). Many macroeconomic transmission mechanisms — particularly second-round wage and price dynamics — operate over longer horizons. The estimated effects likely represent a lower bound on the full impact.
5. **Small cross-section:** With $N = 4$ –5 countries, the Hausman test has limited power. Standard errors are wide, and generalisation beyond this particular group of emerging markets requires caution.

10. Conclusion

This project estimated the macroeconomic impact of the 2026 Gulf oil shock on a panel of emerging net-importing economies, using Pooled OLS, Fixed Effects, and Random Effects panel data models. The key findings are:

- The crisis caused a statistically significant increase in consumer price inflation of approximately 3.1% and a decline in industrial production of approximately 2.2% among emerging importers, confirming the stagflationary transmission channel.
- Fixed Effects is the appropriate estimator, as confirmed by Breusch-Pagan and Hausman tests and the substantial improvement in fit (R^2 gains of 17–36 percentage points over POLS).
- The crisis impact is heterogeneous: emerging importers bear the largest inflation and output costs, while oil exporters and advanced economies are comparatively insulated.
- The nominal effective exchange rate is a key shock absorber: currency appreciation substantially reduces inflationary pass-through, with a NEER elasticity of approximately -0.58 for CPI in the primary panel.
- The identification challenge posed by the global nature of the oil price series was handled by the two-track strategy: POLS uses the continuous oil shock variable; FE/RE use the post-crisis structural break dummy.

These results have clear policy relevance: in the aftermath of geopolitical oil shocks, emerging market policy-makers face a difficult trade-off between inflation control (requiring monetary tightening) and output support (requiring accommodation), with exchange rate management offering one avenue to reduce the severity of the dilemma.

Appendix: R Code Structure Overview

The analysis was carried out in R using the following main packages and workflow:

Step	Description
1. Libraries	<code>plm</code> , <code>lmtest</code> , <code>stargazer</code> , <code>ggplot2</code> , <code>car</code> , <code>sandwich</code> , <code>tidyverse</code>
2. Data loading	<code>Group_Dataset.csv</code> (primary) and <code>panel_dataset_FINAL_merged_fixed.csv</code> (heterogeneity)
3. Transformation	Log CPI/IPI, log NEER, compute <code>oil_pct_change</code> via lag, construct <code>post_crisis</code> dummy, apply ARIMA flags
4. Panel frames	<code>pdata.frame()</code> with <code>index = c("country", "date")</code>
5. Models	<code>plm(..., model = "pooling" / "within" / "random")</code>
6. Diagnostics	<code>phtest()</code> , <code>plmtest()</code> , <code>pbgttest()</code> , <code>bpctest()</code>
7. Robust SEs	<code>coeftest()</code> with <code>vcovHC(..., method = "arellano", type = "HC3")</code>
8. Visualisation	<code>ggplot2</code> figures exported as PNG (9 figures)
9. Heterogeneity	Subsample <code>pdata.frame</code> by <code>oil_balance</code> and <code>group</code> ; repeat FE estimation

All figures were saved as `fig01_brent_price.png` through `fig09_oil_shock_monthly.png`. The script is fully reproducible: loading the two CSV files and running `r_code_fixed.R` end-to-end regenerates all tables and figures.

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